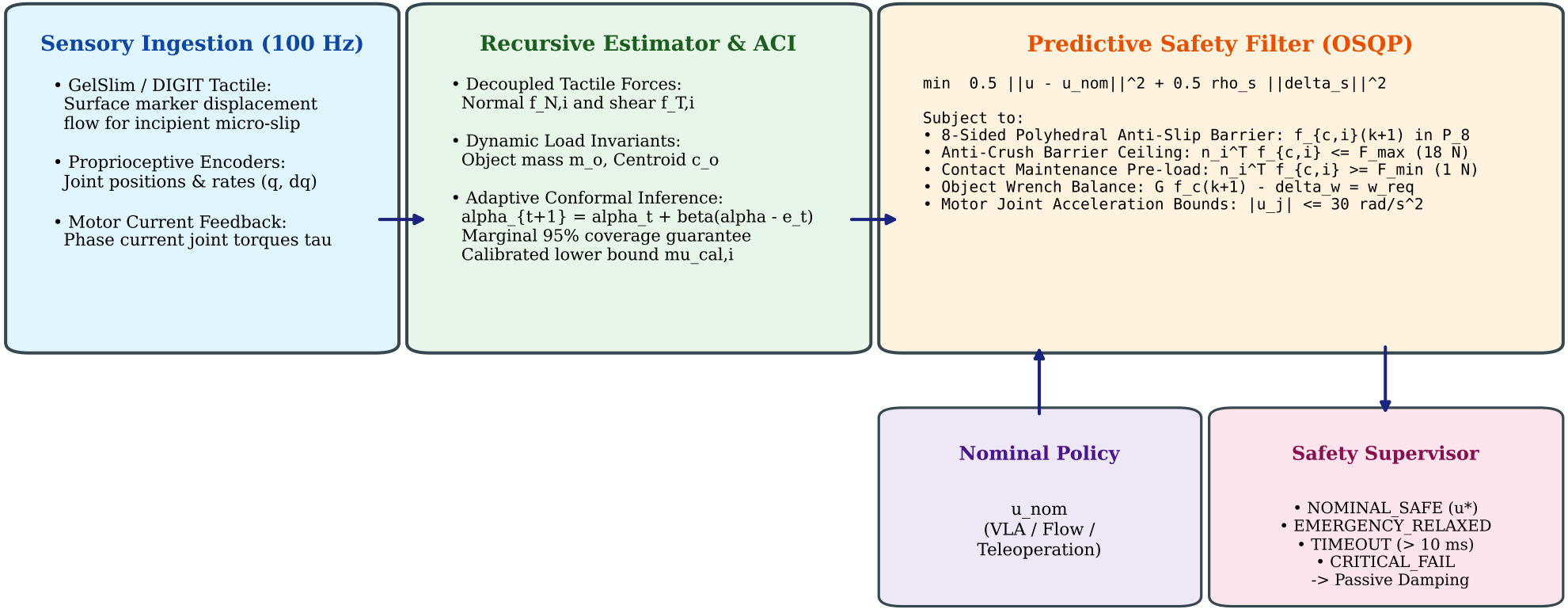

Sensory Ingestion (100 Hz)
• GelSlim / DIGIT Tactile:
Surface marker displacement
flow for incipient micro-slip
• Proprioceptive Encoders:
Joint positions & rates (q, dq)
• Motor Current Feedback:
Phase current joint torques tau
Recursive Estimator & ACI
• Decoupled Tactile Forces:
Normal f_N,i and shear f_T,i
• Dynamic Load Invariants:
Object mass m_o, Centroid c_o
• Adaptive Conformal Inference:
alpha_{t+1} = alpha_t + beta(alpha - e_t)
Marginal 95% coverage guarantee
Calibrated lower bound mu_cal,i
Predictive Safety Filter (OSQP)
min 0.5 ||u - u_nom||^2 + 0.5 rho_s ||delta_s||^2
Subject to:
• 8-Sided Polyhedral Anti-Slip Barrier: f_{c,i}(k+1) in P_8
• Anti-Crush Barrier Ceiling: n_i^T f_{c,i} <= F_max (18 N)
• Contact Maintenance Pre-load: n_i^T f_{c,i} >= F_min (1 N)
• Object Wrench Balance: G f_c(k+1) - delta_w = w_req
• Motor Joint Acceleration Bounds: |u_j| <= 30 rad/s^2
Nominal Policy
u_nom
(VLA / Flow /
Teleoperation)
Safety Supervisor
• NOMINAL_SAFE (u*)
• EMERGENCY_RELAXED
• TIMEOUT (> 10 ms)
• CRITICAL_FAIL
-> Passive Damping